

Extraction of $\Delta 4$ steroids from tooth dentin for LC-MS/MS

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Method Article

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Abstract

Tooth dentin provides a currently underutilized resource for investigating long-term steroid concentrations, and liquid chromatography tandem mass spectrometry (LC-MS/MS) provides a precise and reliable technique for measuring them. This protocol details a procedure developed for extracting Δ^4 steroids from dentin, but it may be used for other solid media as well. Methanol is used to dissolve steroids from powdered dentin samples. After evaporation and reconstitution, supported liquid extraction is used to isolate dissolved Δ^4 steroids for LC-MS/MS analysis. This protocol enables efficient, controlled processing of many samples, and uses readily available materials and supplies. A batch of samples can be processed in about 7 hours. This method requires access to LC-MS/MS and other lab equipment.

Introduction

This protocol was developed in the labs of Metabolism, Endocrinology, and Diabetes in the University of Michigan School of Medicine. It is the result of testing and analyses over the course of one year and hundreds of samples. Results are highly consistent and reproducible when following the protocol. Dentin samples were from tusks of elephants and woolly mammoths as part of a collaboration between faculty in the University of Michigan Museum of Paleontology and in the University of Michigan School of Medicine.

The liquid chromatography tandem mass spectrometry (LC-MS/MS) method used during method development was the internal lab method typically used to measure 24 serum steroids (estriol, 18-hydroxycortisol, cortisol, 18-oxocortisol, cortisone, 18-hydroxycorticosterone, 21-deoxycortisol, aldosterone, estradiol, 11-hydroxytestosterone, 11-ketotestosterone, 11-deoxycortisol, 11-hydroxyandrostenedione, 11-ketoandrostenedione, corticosterone, 16 α -hydroxyprogesterone, estrone, testosterone, 17-hydroxyprogesterone, 11 β -hydroxyprogesterone, 11-ketoprogesterone, androstenedione, 11-deoxycorticosterone, and progesterone). Of the 24 analytes, only testosterone, progesterone, and androstenedione were routinely detected in quantifiable concentrations. Cortisol was also detected, sometimes in high concentrations, but only sporadically.

A dentin-focused LC-MS/MS method could reasonably drop the undetected analytes, but dentin from humans and other species were not analyzed. Elephant serum samples had high concentrations of cortisol, 18-hydroxycorticosterone, cortisone, corticosterone, but only cortisol showed up in dentin samples. Elephant serum also had five other steroids from the 24 measured that were not detected in dentin (18-hydroxycortisol, aldosterone, 11-deoxycortisol, 17-hydroxyprogesterone, 11-deoxycorticosterone), but lacked the other 12. Meanwhile, progesterone was detected in disproportionately high concentrations in elephant and mammoth dentin compared to elephant serum. Although steroid concentrations in dentin should not be considered a direct reflection of serum

concentrations, concentrations of testosterone and androstenedione in dentin proved useful for tracking changes in reproductive physiology during the lives of male elephants and mammoths. The other two analytes (progesterone and cortisol) are the focus on ongoing work. Samples used during method development were typically between 25 and 75 mg and provided clear signals of each analyte, but samples as small as 1 mg had quantifiable concentrations of the three main analytes (testosterone, androstenedione, and progesterone). No samples greater than 100 mg were attempted. Larger samples might require modification to the volume of solvents and sizes of tubes used, but could result in the recovery of other steroids that are present in dentin in lower concentrations.

Reagents

Methanol for extraction and LC-MS/MS (HPLC Grade, JT Baker 9093-33)

Water for extraction and LC-MS (Optima LC/MS Grade Gold Label, Fisher Chemical W6-4)

Methyl *tert*-Butyl Ether for extraction (SLE) (Methyl *t*-Butyl Ether Laboratory Plus, Honeywell LP242-4)

Equipment

Sample vortex mixer (e.g., Fisher Scientific Vortex Mixer)

Sampel rotator (e.g., Thermo Scientific Tube Revolver/Rotator, 88881001)

Microcentrifuge (e.g., Eppendorf Centrifuge 5430)

Positive pressure manifold (e.g., Biotage PRESSURE+ 48)

Vacuum concentrator (e.g., SpeedVac SPD1030)

400 µl Supported Liquid Extraction (SLE) columns or plates (e.g., Biotage ISOLUTE SLE+)

2-mL autosampler vials (e.g., Verex Vials and Caps, Phenomenex AR0-9921-13 and AR0-8952-13-B)

2-mL microcentrifuge tubes (e.g., MP Biomedicals 5076200 and 5067002)

Lysing Matrix D beads (MP Biomedicals 6540-434)

Pipetters and tips (1000 and 200 µl)

Procedure

*Note: An internal standard with known concentrations of deuterated forms of each target analyte in 40% MeOH will be required.

Steroid extraction

1. Label a centrifuge tube for each sample and at least three process blanks.
2. Add Lysing Matrix D beads (~300 mg) to each tube and then record weight of tube + beads.
3. Add powder sample (20-100 mg) to each tube and reweigh to calculate sample mass.
4. Add 1500 µl MeOH + 100 µl internal standard to each tube.
5. Vortex each tube for 60 seconds at 10 krpm and place in rotator.
6. Rotate at 30 rpm for 2 hours at 4°C.

Sample concentration

7. Label an autosampler vial for each sample and process blank.
8. Centrifuge samples for 8 minutes at 14000 g Force.
9. Transfer 1350 µl of supernatant from each tube to the appropriately labeled vial.

*About 1 out of every 20 SLE columns we used had interference for testosterone and androstenedione in our analyses, so backup samples were helpful. If a backup sample/duplicate is desired the supernatant should be divided into 2 vials at this point, but remember that each sample (primary and backup) will contain only half the extracted analyte.

10. Evaporate samples to < 50 µl in vacuum concentrator (90-120 minutes, set to 5.1 torr, ambient temp).
11. Add 40 µl clean MeOH + 300 µl water to each vial.
12. Cap vials and vortex each for 20 seconds at 10 krpm to reconstitute.

Supported liquid extraction (SLE)

*Note: We pre-cooled the rotor from the vacuum concentrator for 3 hours at -20°C prior to use. A unit with refrigeration would make this step unnecessary.

13. Label an autosampler vial for each desired control sample, one SLE process blank, and one double blank.
14. For control samples, add 100 µl control sample + 100 µl internal standard + 200 µl water to each vial.

15. For SLE process blank, add 100 µl internal standard + 300 µl water.
16. For double blank, add 400 µl water.
17. Label an autosampler vial for each sample, process blank, and control sample.
18. Load labeled vials and SLE columns in positive pressure manifold.
19. Pipette each reconstituted sample, control sample, and blank into designated SLE column.
20. Apply 0.5-3.0 psi positive pressure (N₂) to load sample into column (about 5 seconds).
21. Wait 5 minutes.
22. Add 1800 µl Methyl tert-Butyl Ether (MTBE) to each column.
23. Wait 10 minutes to allow passive gravity elution.
24. Apply 0.5-1.0 psi positive pressure (N₂) for 5 seconds.
25. Wait 5 minutes.
26. Blow columns dry with applying pressure incrementally at 0.5psi, 3 psi, 6 psi, and 9 psi (10 seconds each).
27. Move vials with eluted MTBE to vacuum concentrator.
28. Evaporate samples to dryness in vacuum concentrator with precooled rotor (20-30 minutes, set to 5.1 torr, ambient temp).
29. Remove vials from concentrator and place on ice until after reconstituted.
30. Reconstitute samples by adding 200 µl 40% MeOH to each vial.
31. Cap vials and vortex each for 20 seconds at 10 krpm.
32. Store vials in freezer until LC-MS/MS analysis.

Troubleshooting

Time Taken

Steroid extraction from dentin powder using methanol: 3 hours

Concentration and reconstitution to prepare for SLE: 2.5 hours

SLE extraction: 1.5 hours

Anticipated Results

References

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